

TSM 098 | INDEPENDENT PRACTICE

Prime factorization and numerical fractions

While TSM 099 studies Topic 18. Review: Topic 2, Topic 4.

Name: _____ Date: _____

Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Core: prime factorization

1. Write as a product of prime factors.

36

2. Write as a product of prime factors.

48

3. Write as a product of prime factors.

60

4. Write as a product of prime factors.

72

5. Write as a product of prime factors.

84

6. Write as a product of prime factors.

90

7. Write as a product of prime factors.

120

8. Write as a product of prime factors.

180

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Core: simplify using common prime factors

9. Write in lowest terms.

$$\frac{36}{84}$$

10. Write in lowest terms.

$$\frac{48}{180}$$

11. Write in lowest terms.

$$\frac{72}{108}$$

12. Write in lowest terms.

$$\frac{90}{150}$$

13. Write in lowest terms.

$$\frac{126}{210}$$

14. Write in lowest terms.

$$\frac{-84}{132}$$

15. Write in lowest terms.

$$\frac{144}{96}$$

16. Write in lowest terms.

$$\frac{-120}{-168}$$

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Core: equivalent fractions

17. Find the missing numerator.

$$\frac{3}{4} = \frac{?}{24}$$

18. Find the missing numerator.

$$\frac{5}{6} = \frac{?}{30}$$

19. Find the missing numerator.

$$\frac{-2}{9} = \frac{?}{36}$$

20. Find the missing numerator.

$$\frac{7}{10} = \frac{?}{30}$$

21. Find the missing numerator.

$$\frac{4}{15} = \frac{?}{60}$$

22. Find the missing numerator.

$$\frac{11}{12} = \frac{?}{36}$$

23. Find the missing numerator.

$$\frac{-5}{8} = \frac{?}{40}$$

24. Find the missing numerator.

$$\frac{0}{7} = \frac{?}{63}$$

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Extension: compare and explain

25. Are these fractions equivalent? Justify.

$$\frac{6}{8} \text{ and } \frac{9}{12}$$

26. Are these fractions equivalent? Justify.

$$\frac{4}{9} \text{ and } \frac{8}{15}$$

27. Are these fractions equivalent? Justify.

$$\frac{-3}{5} \text{ and } \frac{9}{-15}$$

28. Are these fractions equivalent? Justify.

$$\frac{12}{18} \text{ and } \frac{3}{4}$$

29. Write in lowest terms.

$$\frac{252}{420}$$

30. Write in lowest terms.

$$\frac{-165}{225}$$

31. Explain why dividing only the numerator by 3 changes the value.

$$\frac{9}{12}$$

32. Can multiplying both numerator and denominator by zero produce an equivalent fraction? Explain.